

applications. The Model-View-Controller (or MVC) is software for implementing user interfaces and is the de-facto standard for developing Web applications. Both Tornado and Django provide some form of database support, HTML templating, domain handling, etc.

Tornado vs Django

Even though both Tornado and Django provide the MVC architecture, they are at opposite ends of the spectrum.

Django is more mature and is one among the largest Web-based frameworks for Python. It has a lot of features and takes care of the details for the user, so that you do not need to understand all the basics when making a fully-featured website. It is stable and user friendly too. But Django is very hard to customise and troubleshoot, in some situations, due to its all-encompassing nature. This is one of its drawbacks.

Tornado is lighter in weight and has a few more features than Django. As I said earlier, Tornado is known for its high performance. It runs faster and provides an asynchronous IOLoop for polling requests, and a variety of *oauth* and social media features. It is small in size and runs fast. And it is customisable to the tiniest detail.

Django actually fits most of the platforms. HTTP requests are a fundamental part of every Web application. This applies to Django too. Controllers of this framework allow regular expression matching on URLs, to call the appropriate functions and field incoming requests. It assures sensible default requests and responses, with the ability to override. Tornado is the leanest and meanest among the featured Python Web frameworks. As I mentioned before, Tornado is fast, and is small too—in the sense that it is simple and yet efficient. These features make access points easy for extending the framework itself and directly managing the basic expectations of a Model View Controller framework. It also manages the basic operations of your website, such as parameter parsing, HTTP handling, etc. URLs of both Django and Tornado are almost similar. The only difference is that they use separate classes for each request. This makes it possible to create a class hierarchy for your requests, and then you can paste together a series of pages into a single parent class.

If you want to start a company or business that involves collecting data from users and storing it in a database, Django will really help you out. It has a form generator, which will allow you to define patterns for the database, and it also helps in generating HTML for an application which validates data on submission. The sophisticated data model reads your existing database schema, or creates one for you, and generates code to define the schema as software objects. In Tornado, the database connectivity is not built in by default, but it does not exist in a separate package and is known by the name *torndb*. This database helps in connecting to MySQL and issues your queries. For example, if you want to create your own environment for managing connections and retrying failed queries, then this is the one place you make real use of Tornado. It will work exactly how you want it to.

Django provides the 'View' part of MVC by making use of

its template system efficiently.

The template system syntax of Tornado is a lot like Django's and almost interchangeable: you get template inheritance, variable binding, looping and localisation support just as with Django. Yet, it executes a lot faster with less code under the hood. You can also short-circuit template compilation for speed by carefully instrumenting which files you'll include conditionally. The template values are replaced in Django when you assign variables to an HTML template. This is also possible for the programs written outside of the Python code. This is fairly standard practice these days and Django does a great job of it. The template inheritance is a simple yet powerful way of defining standard headers and footers, as well as other features, so that they exist on every page. The templating is not the fastest on the Web, however; so you should plan for additional processing time.

Since it is such a mature system, Django has all kinds of other features as well: user authentication, localisation, Unicode handling, and the list goes on. If you want a framework that will cover everything you can possibly need and you just don't have the time or skill to do a lot of the work to get it done, then Django is a great system for you. An interesting fact about it is the way it's described—Django is the Web framework for perfectionists with deadlines.

The real differentiating glory of Tornado is the IOLoop, an asynchronous software library that integrates directly with kernel EPoll, KQueue and similar facilities. Also known as an Event Loop, this style of programming allows your program to detach its flow control, so the system can respond to another Web request while you wait for the kernel to dispatch your event. Tornado gives you great access to the internals of the system to make it do exactly what you want. The localisation and Unicode support is pretty thorough - you can create CSV key/value pair files, and look up the values based on the browser language headers. As long as you design your system in an open manner, you can serve your whole website in multiple languages and ensure it executes faster.

I do hope you've got a basic understanding of Python Web frameworks and familiarised yourself with Tornado. Now it's your turn to dive into the code and come up with amazing Web applications. If you're starting a new project, I highly recommend Python. It's a great language with enough clarity, speed and features for you to bring together a global team to build a high-performance website. If you want to choose a Web framework that suits your team and project, the Python community members are there to help you out! They are really awesome! 

References

- [1] <https://devcham.com/articles/16/tornado-examples/>
- [2] <http://carlosgabaldon.com>
- [3] <http://backchannel.org>
- [4] <https://wiki.python.org/main/WebFrameworks>

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